

```
    ],  
    },  
    "id": "ak0224bju1jx"  
  },  
]
```

The key "properties" contains a lot of information about each earthquake ❶. We're mainly interested in the magnitude of each earthquake, associated with the key "mag". We're also interested in the "title" of each event, which provides a nice summary of its magnitude and location ❷.

The key "geometry" helps us understand where the earthquake occurred ❸. We'll need this information to map each event. We can find the longitude ❹ and the latitude ❺ for each earthquake in a list associated with the key "coordinates".

This file contains way more nesting than we'd use in the code we write, so if it looks confusing, don't worry: Python will handle most of the complexity. We'll only be working with one or two nesting levels at a time. We'll start by pulling out a dictionary for each earthquake that was recorded in the 24-hour time period.

NOTE

When we talk about locations, we often say the location's latitude first, followed by its longitude. This convention probably arose because humans discovered latitude long before we developed the concept of longitude. However, many geospatial frameworks list the longitude first and then the latitude, because this corresponds to the (x, y) convention we use in mathematical representations. The GeoJSON format follows the (longitude, latitude) convention. If you use a different framework, it's important to learn what convention that framework follows.

Making a List of All Earthquakes

First, we'll make a list that contains all the information about every earthquake that occurred.

```
eq_explore    from pathlib import Path  
_data.py      import json  
  
# Read data as a string and convert to a Python object.  
path = Path('eq_data/eq_data_1_day_m1.geojson')  
contents = path.read_text()  
all_eq_data = json.loads(contents)  
  
# Examine all earthquakes in the dataset.  
all_eq_dicts = all_eq_data['features']  
print(len(all_eq_dicts))
```

We take the data associated with the key 'features' in the `all_eq_data` dictionary, and assign it to `all_eq_dicts`. We know this file contains records of 160 earthquakes, and the output verifies that we've captured all the earthquakes in the file:

160
